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Assignment 2

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1)

3.17) $\pi(s,a)$ is the given policy

we know: $G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$, $q_\pi(s,a) = E_\pi[G_t \mid S_t = s, A_t = a]$

and $p(s',r \mid s,a) = Pr\{S_{t+1} = s', R_{t+1} = r \mid S_t = s, A_t = a\}$ $\boxed{G_t = R_{t+1} + \gamma G_{t+1}}$

So: $q_\pi(s,a) = E_\pi[G_t \mid S_t = s, A = a] = E_\pi[R_{t+1} + \gamma G_{t+1} \mid S_t = s, A_t = a] =$

$= \sum_{s',r} p(s',r \mid s,a) E_\pi[R_{t+1} + \gamma G_{t+1} \mid S_t = s, A_t = a, S_{t+1} = s', R_{t+1} = r] =$

$= \sum_{s',r} p(s',r \mid s,a) \left(r + \gamma E_\pi[G_{t+1} \mid S_{t+1} = s']\right) =$

$= \sum_{s',r} p(s',r \mid s,a) \left(r + \gamma \sum_{a'} Pr\{A_{t+1} = a' \mid S_{t+1} = s'\} E_\pi[G_{t+1} \mid S_{t+1}, A_{t+1} = a']\right) =$

$= \boxed{\sum_{s',r} p(s',r \mid s,a) \left(r + \gamma \sum_{a'} \pi(s',a') q_\pi(s',a')\right)}$

1)

3.19)

we know: $V_\pi(s) = E_\pi(G_t \mid S_t = s)$, $q_\pi(s,a) = E_\pi[G_t \mid S_t = s, A_t = a]$

$$q_\pi(s,a) = E_\pi[G_t \mid S_t = s, A_t = a] =$$

$$= E_\pi[R_{t+1} + \gamma G_{t+1} \mid S_t = s, A_t = a] = E[R_{t+1} + \gamma E_\pi[G_{t+1} \mid S_{t+1}] \mid S_t = s, A_t = a]$$

$$= \boxed{E[R_{t+1} + \gamma V_\pi(S_{t+1}) \mid S_t = s, A_t = a]} =$$

$$= \boxed{\sum_{s',r} p(s',r \mid s,a)\left(r + \gamma V_\pi(s')\right)}$$ → na expected form